

Integrating Impairments in Reaction Time and Executive Function Using a Diffusion Model Framework

Sarah L. Karalunas · Cynthia L. Huang-Pollock

Published online: 20 January 2013
© Springer Science+Business Media New York 2013

Abstract Using Ratcliff's diffusion model and ex-Gaussian decomposition, we directly evaluate the role individual differences in reaction time (RT) distribution components play in the prediction of inhibitory control and working memory (WM) capacity in children with and without ADHD. Children with ($n=91$, $\bar{x}$ age=10.2 years, 67 % male) and without ADHD ($n=62$, $\bar{x}$ age=10.6 years, 46 % male) completed four tasks of WM and a stop signal reaction time (SSRT) task. Children with ADHD had smaller WM capacities and less efficient inhibitory control. Diffusion model analyses revealed that children with ADHD had slower drift rates (v) and faster non-decision times (Ter), but there were no group differences in boundary separations (a). Similarly, using an ex-Gaussian approach, children with ADHD had larger τ values than non-ADHD controls, but did not differ in μ or σ distribution components. Drift rate mediated the association between ADHD status and performance on both inhibitory control and WM capacity. τ also mediated the ADHD-executive function impairment associations; however, models were a poorer fit to the data. Impaired performance on RT and executive functioning tasks has long been associated with childhood ADHD. Both are believed to be important cognitive mechanisms to the disorder. We demonstrate here that drift rate, or the speed at which information accumulates towards a decision, is able to explain both.

Keywords ADHD · Executive function · Reaction time · Ex-Gaussian · Diffusion model

S. L. Karalunas (✉)
ADHD Research Study UHN80R1, Oregon Health Sciences
University, 3181 SW Sam Jackson Park Rd,
Portland, OR 97239-9979, USA
e-mail: karaluna@ohsu.edu

C. L. Huang-Pollock
Department of Psychology, The Pennsylvania State University,
0113 Moore Bldg,
University Park, PA 16801, USA

When speeded responses are required, children with ADHD often respond less accurately, are slower, and have more variable reaction times (RTs) compared to their same-aged peers (Castellanos et al. 2005; Douglas 1999; Hervey et al. 2006; Kuntsi and Stevenson 2001; Leth-Steensen et al. 2000). Like ADHD itself (Faraone et al. 2005), RT and RT variability are also highly heritable (Kuntsi et al. 2006; Kuntsi and Stevenson 2001), leading some to suggest that these performance characteristics may be important cognitive mechanisms in the disorder that may elucidate underlying neural dysfunction (Castellanos and Tannock 2002). In addition, slow RTs, as an index of information processing speed, have been hypothesized to explain ADHD-related deficits on a variety of executive tasks (Alderson et al. 2007; Lijffijt et al. 2005), which has implications for characterization of the ADHD cognitive phenotype and search for underlying mechanisms.

That being said, although slower RTs among children with ADHD is a common finding, it is certainly not ubiquitous, and there is significant between-study heterogeneity in the presence and size of group differences for mean RTs (MRT, Frazier et al. 2004; Huang-Pollock et al. 2012). Mechanistic interpretation of the presence or absence of group differences in RTs, either at the cognitive or neural levels, is also limited because RTs are influenced by multiple interacting processes. These include, but are not limited to, stimulus encoding, rate of information processing, motor preparation and output, speed-accuracy trade-off effects, and response bias (Ratcliff 2002). All have been hypothesized to be involved in ADHD (Banaschewski et al. 2008; Halperin and Schulz 2006; Hurks et al. 2005; Klimkeit et al. 2005; Lajoie et al. 2005; Sergeant 2000; Sergeant and Scholten 1985; Steger et al. 2001). Thus, modeling these processes and predicting how they ultimately produce a RT is critical not only to explaining the presence or absence of group differences in MRT, but also to understanding why speed of response for even simple judgments is so

commonly associated with performance on higher level cognitive constructs (e.g. intellectual and executive functions [EF], Grudnik and Kranzler 2001; Salthouse et al. 1999; Schmiedek et al. 2007), and to developing a more comprehensive theory of why poor performance on tasks of EF and speeded processing are often observed in ADHD.

The importance of better understanding mechanisms contributing to group differences in RTs has recently lead to the use of alternative methods of characterizing RTs in ADHD, such as ex-Gaussian approaches (Buzy et al. 2009; Epstein et al. 2011; Geurts et al. 2008; Leth-Steensen et al. 2000). Ex-Gaussian analyses provide separate estimates of the mean (μ) and standard deviation (σ) of the Gaussian portion of the RT distribution, and the mean/standard deviation of the exponential portion of the distribution (τ). The most consistent significant ADHD-control group differences are reported in the exponential upper tail of the RT distribution (τ), suggesting that children with ADHD have a large number of excessively long RTs relative to non-ADHD controls (Buzy et al. 2009; Epstein et al. 2011; Leth-Steensen et al. 2000). It may be that group differences in the upper-tail of the distribution account for group differences in mean or standard deviations of RTs, however, some studies continue to note group differences in both μ and σ in addition to groups differences in τ (Buzy et al. 2009; Epstein et al. 2011). Although the patterns of statistical significance are less consistent for μ and σ , a relatively small number of studies have applied ex-Gaussian analyses, leaving questions about which parameters best differentiate children with ADHD from non-ADHD controls.

Even if we accept that the largest group differences are seen in the exponential tail of the RT distribution, cognitive interpretations of ex-Gaussian parameters remain uncertain. τ has sometimes been interpreted as reflecting higher-order processing (e.g. attention lapses, Leth-Steensen et al. 2000) while μ and σ relate more strongly to motor responding; however, the opposite interpretations have also been made (for review see Matzke and Wagenmakers 2009). A significant limitation of the ex-Gaussian approach that may contribute to heterogeneity in findings and difficulty with interpretation of parameters is that they do not take into account response accuracy, which prevents direct assessment of speed-accuracy trade-off effects. Prioritizing speed over accuracy has often been noted as a characteristic of performance for children with ADHD (Mulder et al. 2010; Sergeant et al. 1999), and the degree to which this occurs confounds the interpretation of RT data (Matzke and Wagenmakers 2009).

Analyses based on a drift diffusion model of RTs (Ratcliff and Rouder 1998) provide one way of statistically disentangling the multiple factors influencing RTs. Widely used in the cognitive psychology literature (Balota and Yap 2011; Kühn et al. 2011; Ratcliff et al. 2004; Schmiedek et al. 2009;

Spaniol and Bayen 2005; Thapar et al. 2003), diffusion models have been developed to explain decision-making in forced-choice paradigms for which relatively rapid (~ 1 s) response decisions are required (Ratcliff and Rouder 1998). These models assume that information about a stimulus is accumulated via a noisy information accumulation process until a decision criterion is met, at which point a response is initiated. That is, when faced with a forced-choice decision, participants are assumed to set criteria for how much information they require to commit to either of the response options. Individuals who require less information will respond more quickly but will also make more errors than those who require more information. The process is described as “noisy” because random neural activity unrelated to the decision process impacts the speed and efficiency with which a person is able to accumulate information relevant to the decision itself. Finally, processes that are not directly related to the response decision, such as stimulus encoding and motor preparation, will also impact the final RT.

Practically, the diffusion model provides estimates of: 1) *drift rate* (v ; an index of how quickly and efficiently an individual can accumulate information to inform their response decision, which is theoretically linked to neural signal:noise ratio); 2) *boundary separation* (a ; how “sure” a person needs to be before committing to a response, or their speed-accuracy trade-off setting); and 3) *non-decision time* (T_{er} ; the time it takes to complete all other information processes not involved in stimulus discrimination, e.g., encoding, memory access, motor preparation; Ratcliff and McKoon 2008; see Fig. 1 and Table 1).

Although not the only model for understanding RTs (for review see Bogacz et al. 2006), diffusion models are well-validated and show strong convergent and divergent validity, such that manipulations of stimulus clarity uniquely impact drift rate and manipulations of the complexity of motor response required uniquely impact non-decision time (Voss et al. 2004). Further, recent studies in both humans and non-human primates have demonstrated correspondence between diffusion model parameters and real-time functioning of neural networks (Beck et al. 2008; Philiastides et al. 2006; Philiastides and Sajda 2006; Ratcliff et al. 2003; Ratcliff et al. 2009). Thus, application of diffusion models has particular promise for understanding the processes underlying RTs and for elucidating connections between underlying neurological dysfunction and performance on cognitive tasks.

In addition to group differences in RT distributions, there is robust evidence that children with ADHD perform more poorly on tasks of response inhibition and working memory (WM) relative to non-ADHD controls (Lijffijt et al. 2005; Martinussen et al. 2005; Willcutt et al. 2005). However, at a mechanistic level how deficits in inhibition and WM may be

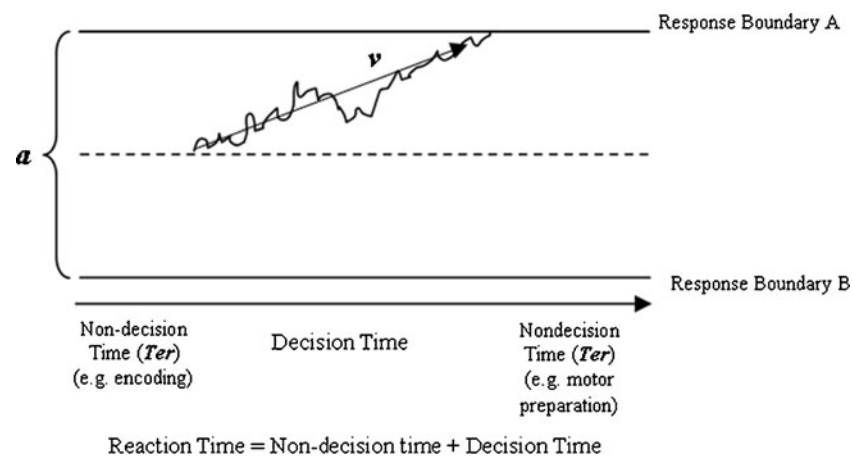


Fig. 1 Diffusion model parameters depicted for a hypothetical single trial (adapted from Ratcliff and Rouder 1998). Drift rate (v) is the rate at which information accumulates towards a decision boundary, determined by speed of information processing and “noise” unrelated to the

decision processes; boundary separation (a) indicates the conservativeness of the response criterion with wider separations indicating more conservative responding; and non-decision time (Ter) includes all non-decision processes, such as stimulus encoding and motor preparation

substantively related to RT variables remains unclear. In the case of response inhibition, children with ADHD have slower stop signal reaction times (SSRTs) than non-ADHD controls but also tend to have slower MRTs (Lijffijt et al. 2005). This lack of specificity has led to suggestions that group differences in SSRT are a result of a general cognitive deficit, such as slow or inefficient information processing, rather than a specific deficit in inhibition (Alderson et al. 2008; Lijffijt et al. 2005). Such a finding suggests that of the RT components, drift rate (v) would be uniquely associated with SSRT. However, problems with motor preparation and output (Klimkeit et al. 2005; Steger et al. 2001) or strategic adjustments to response criterion that maximize both go-speed and stopping accuracy (Verbruggen and Logan 2009) could also result in similar effect sizes for group differences in MRT and SSRT.

Regarding WM, some of the strongest empirical evidence suggests that a large proportion of the developmental improvement in WM span can be attributed to increased

processing speed (Fry and Hale 2000; Kail 2007), suggesting that drift rate may uniquely mediate WM capacity. Consistent with this possibility, Schmiedek et al. (2007) found that in young adults both speed-accuracy trade-offs (boundary separation) and information processing speed/efficiency (drift rate) were related to WM capacity, but that the strength of the relationship for drift rate was nearly twice that of boundary separation. RT decomposition approaches have not been applied to this question within the context of ADHD, however using a latent variable approach, Huang-Pollock et al. (2011) recently found that processing speed accounted for a significant proportion of variance in WM span among children with and without ADHD; however, which RT component(s) are most critically responsible for that relationship is not known.

The current study first examines group differences in the ex-Gaussian parameters that descriptively characterize the RT distribution of children with ADHD. We then apply diffusion model analyses to compare groups on parameters

Table 1 Description of diffusion model parameters

Parameter	Interpretation	Additional explanation
Drift rate (v)	Speed/efficiency of information processing	The average slope of the line reflects the rate of information processing. Deviations from the average slope indicate the effects of random neural noise on the decision process. Larger values of v indicate faster and more efficient information processing.
Boundary separation (a)	Speed-accuracy trade-off setting	The more conservative the response criterion, the farther apart the response boundaries are. Thus, larger values of a indicate greater emphasis on accuracy than speed in the decision processes
Non-decision time (Ter)	Time for all non-decisional processes	Ter processes prior to the start of the decision process (e.g. stimulus encoding) and following the response decision (e.g. motor response preparation and execution). Larger values of Ter indicate longer non-decisional processing times

with well-validated cognitive interpretations. Finally, we determine which RT distributional components may help explain the relationship between ADHD status and poor EF.

Methods

Participants

Children between the ages of 8–12 with ADHD ($\bar{x}$ age = 10.2 years, 67 % male) and without ADHD ($\bar{x}$ age = 10.6 years, 46 % male) were community recruited from

local schools, newspaper/radio ads, and distributed flyers in the Central, York, and Dauphin county areas of Pennsylvania as part of a study approved by the Pennsylvania State University Institutional Review Board (#32136). ~80 % of both the ADHD and non-ADHD samples identified as Caucasian and non-Hispanic (82.8 % and 79.1 %, respectively). Both ADHD and non-ADHD groups reported a median annual household income of \$61,000–\$71,000. A full description of the final sample can be found in Table 2.

In an initial screening phase, children were screened as *potentially* in the ADHD group if both parent and teacher

Table 2 Sample characteristics and task performance measures

	Control	ADHD	Group comparison
Basic demographics			
N	62	91	
(Boys:Girls)	29:33	61:30	$\chi^2(1)=6.25, p=0.012$
Subtype (C,I,H)		40,48,3	
Full scale IQ	106.94 (10.40)	104.42 (11.73)	$F(1,151)=1.86, p=0.175, \eta^2=0.01$
% Caucasian & non-Hispanic	79.1	82.8	$\chi^2(1)=1.71, p=0.191$
Median annual household income	\$61,000–\$70,000	\$61,000–\$70,000	$Z=-1.24, p=0.216$
Age (years)	10.56 (1.21)	10.19 (1.39)	$F(1,151)=2.87, p=0.092, \eta^2=0.02$
Total #Att symptoms	0.79 (1.01)	8.05 (1.21)	$F(1,151)=1522.61, p<0.001, \eta^2=0.91$
Total #HI symptoms	0.29 (0.52)	4.99 (3.01)	$F(1,151)=147.09, p<0.001, \eta^2=0.49$
ODD (<i>n</i>)	1	30	$\chi^2(1)=23.07, p<0.001$
CD (<i>n</i>)	0	6	$\chi^2(1)=4.35, p=0.037$
GAD (<i>n</i>)	1	6	$\chi^2(1)=2.17, p=0.140$
MDD/Dysthymia (<i>n</i>)	0	1	$\chi^2(1)=0.73, p=0.391$
Working memory			
Letter span	0.29 (0.14)	0.20 (0.08)	$F(1,146)=20.44, p<0.001, \eta^2=0.12$
Spatial span	0.26 (0.10)	0.21 (0.09)	$F(1,145)=9.76, p=0.002, \eta^2=0.06$
Digits backwards	0.27 (0.10)	0.24 (0.10)	$F(1,148)=4.01, p=0.05, \eta^2=0.03$
Finger windows backwards	0.36 (0.14)	0.28 (0.13)	$F(1,143)=12.62, p=0.001, \eta^2=0.08$
Stop task			
% Choice accuracy	94.1 (7.96)	86.1 (10.43)	$F(1,151)=26.07, p<0.001, \eta^2=0.15$
# Omissions	3.10 (4.04)	6.33 (5.41)	$F(1,151)=16.00, p=0.001, \eta^2=0.10$
Mean RT	749.97 (174.66)	758.04 (123.71)	$F(1,151)=0.11, p=0.738, \eta^2<0.01$
SD RT	196.50 (61.82)	231.88 (52.33)	$F(1,151)=14.54, p=0.001, \eta^2=0.09$
Stop signal RT	285.73 (90.02)	422.51 (152.18)	$F(1,151)=40.40, p<0.001, \eta^2=0.21$
Ex-Gaussian components			
Mu (μ)	596.60 (182.64)	564.55 (128.59)	$F(1,151)=1.62, p=0.205, \eta^2=0.01$
Sigma (σ)	113.92 (59.12)	126.86 (56.74)	$F(1,151)=1.85, p=0.175, \eta^2=0.01$
Tau (τ)	148.36 (75.63)	185.42 (72.95)	$F(1,151)=9.24, p=0.003, \eta^2=0.06$
Diffusion modeling components			
Drift rate (v)	-3.14 (0.85)	-2.53 (0.67)	$F(1,151)=23.86, p<0.001, \eta^2=0.14$
Boundary separation (a)	1.72 (0.33)	1.65 (0.34)	$F(1,151)=2.05, p=0.154, \eta^2=0.01$
Non-decision time (Ter)	0.46 (0.13)	0.42 (0.09)	$F(1,151)=5.17, p=0.024, \eta^2=0.03$

Att Attention, *HI* Hyperactive/Impulsive, *ODD* oppositional defiant disorder, *CD* conduct disorder, *GAD* Generalized Anxiety Disorder, *MDD* Major Depressive Disorder, *RT* Reaction time, *SD* Standard Deviation. Smaller absolute values of v indicate slower drift rates. Smaller values of a indicate greater speed-accuracy trade off, and *Ter* is reported in seconds

report exceed the 85th percentile (T-score >61) on either the (a) Behavioral Assessment Scale for Children (BASC-2: Reynolds and Kamphaus 2004) Attention or Hyperactivity problem subscales or (b) Conners' Rating Scales—Revised (Conners': Conners 2001) Inattention, Hyperactivity, or ADHD subscales. To be included in the final ADHD group, children were required to meet criteria for DSM-IV ADHD by parent report on the National Institute of Mental Health's Diagnostic Interview Schedule of Childhood for DSM-IV (DISC-4) including age of onset, impairment, and cross-situational severity. Using the “or” algorithm (Lahey et al. 1994) to integrate parent (DISC-IV) and teacher (ADHD-RS) report of specific symptoms, 40 children were identified as combined subtype, 48 with the primarily inattentive subtype, and 3 with the primarily hyperactive subtype. Thirty one children were prescribed a psychostimulant medication, and they were required to be medication free for at least 24 h prior to the day of testing (mode >48 h). Children taking non-stimulant medications were excluded from the study.

Non-ADHD controls did not meet criteria for ADHD on the DISC-IV. In addition, they were required to be below the 79th percentile (T-score ≤58) on all of the above listed rating scales and indices and must never have been diagnosed or treated for ADHD. Common childhood disorders (i.e., anxiety, depression, oppositional defiant disorder, and conduct disorder) were identified using the DISC-4 (see Table 2), but the presence of these disorders was not exclusionary for either ADHD or controls. Children with estimated IQ <80 were excluded.

Procedures

Prior to participation, informed written consent from parents and verbal assent from children were obtained. Parents were given monetary compensation and informal feedback regarding relevant clinical findings. Children were given a small prize worth <\$2. All children completed the cognitive screening measures and the experimental paradigms in the same order as part of a larger test battery that took place in two, three-hour long testing sessions.

Cognitive Screening Measures A 2-subtest short form (Vocabulary, Matrix Reasoning) of the Wechsler Intelligence Scale for Children—IV (WISC-IV: Wechsler 2003) provided an estimated IQ for 94 (61 %) participants. Another 59 (39 %) participants completed a 4 subtest short form (Vocabulary, Matrix Reasoning, Arithmetic, Symbol Search). The correlation of the 2 and 4 subtest short forms with the full 12-subtest battery are 0.87 and 0.93, respectively (Sattler 2008).

Working Memory The Letter Span and Spatial Span tasks were based in part on the Eprime programs provided by

Engle and colleagues (Conway et al. 2005). In the *Letter Span Task*, children sequentially viewed a letter of the alphabet at a pace of 1 per second. The number of to-be-remembered items increased from two to nine, and after all items were presented, children were asked to recall them in the order they were presented. Three items were presented per set size, and the task was discontinued if children failed all items of a set size. In the *Spatial Span Task*, children viewed a 4×4 grid in which one square randomly turned red. As with the Letter Span task, the number of to-be-remembered items increased from two to nine, and after all items were presented, children were asked to recall them in the order they were presented. Three items were presented per set size, and the task was discontinued if children failed all items of a set size.

We also administered Digits Backwards from the WISC-IV (WISC-IV: Wechsler 2003), and a novel Finger Windows Backwards test that was adapted from the Finger Windows Forwards subtest of the WRAML-2 (Sheslow and Adams 2003). In the *Digits Backwards* task, children listened to a research assistant read aloud a list of numbers at the pace of one per second. They were then asked to repeat the digits out loud in the correct backwards sequence. Two items were presented per set size and the task was discontinued if children failed both items of a set size. In the *Finger Windows Backwards* condition, children watched a research assistant place the tip of a pen through round holes or “windows” in an opaque plastic board one at a time, at the pace of one per second. Children were then asked to place their finger in the holes in the correct backwards sequence. The task was discontinued following three consecutive errors.

All WM tasks were scored using the “all or nothing” load scoring approach (Conway et al. 2005), in which children received credit for the total number of to-be-recalled targets they recalled in the correct sequential order. Scores are reported as a proportion of the total number correct out of total possible correct. To compute reliability, three separate total scores were calculated for each child using the first, second, and third trials (or first and second trials if a third trial was not provided, as in Digit Span) at each span length. Chronbach's α among those span scores was then calculated to determine whether performance was consistent across trials. α 's were 0.85 for Letter Span, 0.77 for Spatial Span, 0.67 for Digits Backwards, and 0.77 for Finger Windows Backwards.

The amount of missing data was slight (2 % for digits backwards, 3.3 % for letter span, 3.9 % for spatial span, and 5.2 % for finger windows backwards) and missing completely at random, χ^2 (38, $N=153$)=33.26, $p=0.69$; they were therefore replaced using single EM imputation (Widaman 2006).

Stop Signal Reaction Time (SSRT) Task Children were administered a 200-trial tracking version of the Logan Stopping Task. Children completed 20 practice trials to ensure understanding of the directions. The task was then administered in 5 blocks of 40 trials with an optional rest break between blocks. For each trial, a central fixation point appeared for 200 ms. On 75 % of trials (i.e. “go” trials), an “X” or an “O” was presented in the center of the screen for 1,000 ms, and children made a forced-choice response whether an “X” or an “O” had appeared. Children were given 2,300 ms to respond, after which the next trial automatically commenced. On 25 % of trials (i.e. 50 “stop” trials), an auditory tone was presented to indicate that they should not respond. An initial MRT was determined based on the practice trials and the auditory stop tone was initially set to occur 250 ms before the MRT. The MRT was then dynamically recalculated after each correct go trial and the delay at which the stop tone was presented was adjusted dynamically in 50 ms increments to maintain an overall ~50 % accuracy rate. (In our sample, non-ADHD controls were able to maintain a slightly higher rate of successful inhibition than children with ADHD [57 % vs. 52 %, $p < 0.001$]). SSRT, the amount of time a child needs in order to successfully inhibit a response 50 % of the time, was calculated by subtracting the mean delay from the child’s MRT. Reliability (α) across the five blocks of trials was 0.82 (number of hits), 0.95 (MRT), 0.85 (standard deviation of RT; SDRT), and 0.88 (SSRT).

Only children who were able to maintain >70 % accuracy rate on the “go” trials were included in analyses. This resulted in 9 children with ADHD being excluded. An additional 2 non-ADHD controls and 3 children with ADHD were excluded because they did not make any errors and so diffusion model error distributions could not be fitted to the data. Included children with ADHD did not differ from excluded children with ADHD in age, estimated IQ, number of attention symptoms, or number of hyperactive symptoms (all $p < 0.287$, all $\eta^2 < 0.01$). All results (including the sample description in Table 2) are presented with these 14 children excluded.

Although primarily used as a measure of response inhibition in the ADHD literature, the stop task has been conceptualized as consisting of two partially independent tasks—the “go” task, in which the child needs to decide whether an “X” or “O” was presented and respond appropriately, and the “stop” task, in which the child needs to inhibit the “go” process on the minority of trials. It is unclear how group differences in inhibitory control might affect the RT distribution (Schachar et al. 2004; Verbruggen et al. 2008). In our sample, RTs to the go trials immediately preceded by a stop signal trial (referred to as stop + 1 trials) were slower than RTs to go trials not immediately preceded by a stop trial ($p = 0.001$, $\eta^2 = 0.07$), but stop + 1 trial slowing did not differ between groups, ($p = 0.198$). Although there was no group difference in the effect, to try

to minimize the effect of the stop task context on estimation of diffusion and ex-Gaussian RT indices, stop + 1 trials were removed from the RT series. Anticipation RTs <150 ms, which accounted for less than 0.25 % of the data, were also removed.

Ex-Gaussian parameters μ , σ , and τ were computed using the egfit tool in MATLAB. (Lacouture and Cousineau 2008). This function performs an iterative search process using a simplex method to compare the observed RT distribution to an ex-Gaussian probability density function (PDF) with known values of μ , σ , and τ . In each iteration, the parameter values of the PDF are adjusted until maximum fit to the observed data is achieved. Because ex-Gaussian models do not take into account response accuracy, these parameter estimates were obtained from the distribution of correct RTs only (Epstein et al. 2011; Hervey et al. 2006; Leth-Steensen et al. 2000). This resulted in a *minimum* of 72 trials available for analysis for each participant, consistent with the number of trials used in previous studies of ADHD (Buzy et al. 2009; Geurts et al. 2008; Leth-Steensen et al. 2000).

Diffusion model parameters were estimated using the Fast-dm modeling technique (Voss and Voss 2007). The Fast-dm program also uses an iterative distribution fitting approach with downhill-simplex method to compare the observed RT distribution to the distribution predicted to occur with specific values of v , a , and Ter . Thus, similar to the ex-Gaussian estimation process, parameters are derived via a multi-dimensional space search in which distributions with known diffusion parameters are compared to observed data until the maximum fit between the predicted and observed distributions is achieved. Full details of the estimation procedure are not described here due to space constraints but can be found in Voss et al. 2004 and Voss and Voss 2007. Because both speed and accuracy are accounted for in the diffusion model, both the correct and error RTs were used, reflecting the upper and lower boundaries of the model, respectively. FastDM can reasonably recover parameter estimates with as few as 20 trials (Voss and Voss 2007). Participants for whom the diffusion model was not able to adequately fit the data would have been excluded from analyses; however, no participants needed to be excluded for this reason. Groups did not differ in model fit with both groups’ p values approaching 1 ($p = 0.96$ and 0.98 for the non-ADHD and ADHD groups, respectively). Smaller absolute values of v indicate slower drift rates. Smaller values of a indicate greater speed-accuracy trade off, and Ter is reported in seconds.

Results

Sample Characteristics

In general, no subtype differences were observed (the single exception is noted below) and so ADHD subtypes were

collapsed for comparison with controls. Group differences on parent and teacher standardized ratings were consistent with the diagnostic groupings (all $\eta^2 > 0.42$, all $p < 0.001$; see Table 2). Children with ADHD did not differ from non-ADHD controls in their IQ or age (all $\eta^2 < 0.02$, all $p > 0.092$), but the ADHD group had a greater proportion of boys ($\chi^2[1] = 6.25$, $p = 0.012$). Results for the group differences reported below were all unchanged when gender, estimated IQ, and age were included in the model as covariates, and so results reported are without those covariates in the model.

Group Differences on Traditional Task Performance Measures

On the go-trials of the stop task, children with ADHD were less accurate and made more omissions than their non-ADHD peers (both $p < 0.001$). Groups did not differ in their MRTs for correct go trials ($F[1,151] = 0.11$, $p = 0.738$, $\eta^2 < 0.01$), but children with ADHD did have larger standard deviation of RTs than non-ADHD controls ($F[1,151] = 14.54$, $p = 0.001$, $\eta^2 = 0.09$). Children with ADHD also had slower SSRTs than non-ADHD controls ($F[1,151] = 40.40$, $p < 0.001$, $\eta^2 = 0.21$). There was a subtype difference for SSRT such that children with ADHD-C had significantly worse SSRTs than children with ADHD-I ($p = 0.01$). Finally, children with ADHD had worse performance on all of the WM tasks than controls (all $p < 0.05$; see Table 2 for full description of task performance).

Ex-Gaussian Analyses and Diffusion Model Analyses

When groups were compared on ex-Gaussian parameters, children with ADHD did not differ from non-ADHD controls in either μ ($F[1,151] = 1.62$, $p = 0.205$, $\eta^2 = 0.01$) or σ ($F[1,151] = 1.85$, $p = 0.175$, $\eta^2 = 0.01$). Children with ADHD

did have larger values for τ than non-ADHD controls ($F[1,151] = 9.24$, $p = 0.003$, $\eta^2 = 0.06$), indicating that increased SDRT was a result of a larger number of RTs in the exponential upper tail of the distribution.

To clarify the specific cognitive processes that underlie group differences in RT distributions, diffusion model analyses were also applied. Children with ADHD had slower drift rates than non-ADHD controls ($F[1,151] = 23.86$, $p < 0.001$, $\eta^2 = 0.14$) and faster non-decision times ($F[1,151] = 5.17$, $p = 0.024$, $\eta^2 = 0.03$). However, groups did not differ in their speed-accuracy trade-offs ($F[1,151] = 2.70$, $p = 0.102$, $\eta^2 = 0.02$; see Table 2 for group means and standard deviations). Correlations between the stop task, WM, ex-Gaussian, and diffusion model parameters can be found in Table 3.

Mediation Analyses

To determine which diffusion model RT components were associated with and might account for ADHD-related deficits in inhibitory control, we conducted a multiple mediator regression analysis (Preacher and Hayes 2008). Model paths reflecting group differences (which are described above) are not discussed again, but can be found in Figs. 2 and 3. Slower drift rates were associated with worse inhibitory control ($p = 0.005$); this was not the case with either a or Ter (both $p > 0.46$). The Sobel test of the indirect effect of ADHD status on inhibitory control through drift rate was significant (total unstandardized indirect effect = 35.61, $SE = 10.86$, $p = 0.001$), indicating that drift rate mediated the relationship between ADHD status and inhibitory control, accounting for 24 % of the variance in the path from ADHD status to SSRT. The indirect effects for a (total unstandardized indirect effect = 1.30, $SE = 2.57$, $p = 0.61$) and Ter (total unstandardized indirect effect = 3.02, $SE = 4.20$, $p = 0.47$)

Table 3 Correlation matrix for ex-Gaussian, diffusion model, working memory, and stop task outcomes

	1	2	3	4	5	6	7	8	9	10	11
	μ	σ	τ	ν	a	Ter	SSRT	LSpan	DSB	Sspan	FWB
1	1										
2	0.73***	1									
3	-0.36***	-0.21**	1								
4	-0.28***	0.03	0.64***	1							
5	0.37***	0.25**	0.25**	-0.22**	1						
6	0.85***	0.38***	-0.40***	-0.38***	0.13	1					
7	-0.16*	0.09	0.42***	0.44***	-0.15	-0.23**	1				
8	0.20**	0.04	-0.24**	-0.32***	0.10	0.17*	-0.21**	1			
9	-0.03	-0.10	-0.15	-0.07	-0.13	-0.03	-0.12	0.30***	1		
10	-0.04	-0.15	-0.26**	-0.32***	-0.01	0.07	-0.27***	0.37***	0.25**	1	
11	-0.10	-0.07	-0.24**	-0.31***	0.03	0.04	-0.39***	0.48***	0.26***	0.51***	1

SSRT Stop Signal Reaction Time Task, LSpan Letter Span, DSB Digit Span Backwards, Sspan Spatial Span, FWB Finger Windows Backwards, *** $p \leq 0.001$, ** $p \leq 0.01$, * $p < 0.05$

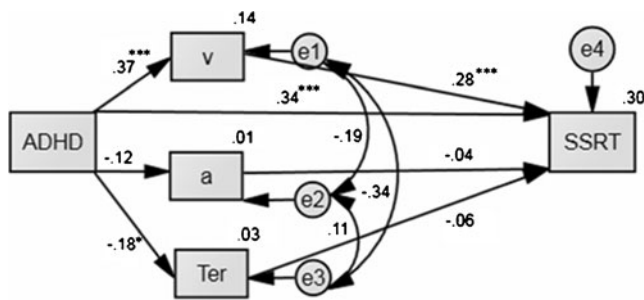


Fig. 2 Multiple mediation model demonstrating the relationships between ADHD, RT components, and stop signal reaction time (SSRT). Standardized parameter estimates displayed. *** $p < 0.001$, * $p < 0.05$

were not significant. A significant direct path from ADHD status to SSRT was also observed ($p < 0.001$; for full model see Fig. 2).

To determine which diffusion model RT components were associated with and might account for WM deficits in ADHD, we conducted an SEM analysis with the three primary diffusion model variables as mediators. The error residuals for those parameters were allowed to correlate (Preacher and Hayes 2008). The four WM tasks loaded onto a single domain general WM construct. When this latent WM factor was set as the dependent variable, the fit of the model was very good, $\chi^2[14] = 17.69$, $p = 0.22$, CFI = 0.98, RMSEA = 0.04. Only the parameter estimate for v was significantly associated with WM capacity ($p = 0.02$), in which faster drift rates predicted greater WM capacity. The Sobel test of the indirect effect of ADHD status on WM capacity through drift rate was significant (total unstandardized indirect effect = -0.01 , $SE = 0.01$, $p = 0.02$) indicating that drift rate mediated the relationship between ADHD status and WM capacity, accounting for 21 % of the variance in the path from ADHD status to WM capacity. The indirect effects for a (total unstandardized indirect effect = 0.01 , $p = 0.36$) and Ter (total unstandardized indirect effect < 0.001 , $p = 0.48$) were not significant and neither could be considered mediators. As with findings for SSRT, a significant direct path from ADHD to WM was also observed ($p < 0.001$; for full model see Fig. 3), and the standardized direct effect of ADHD

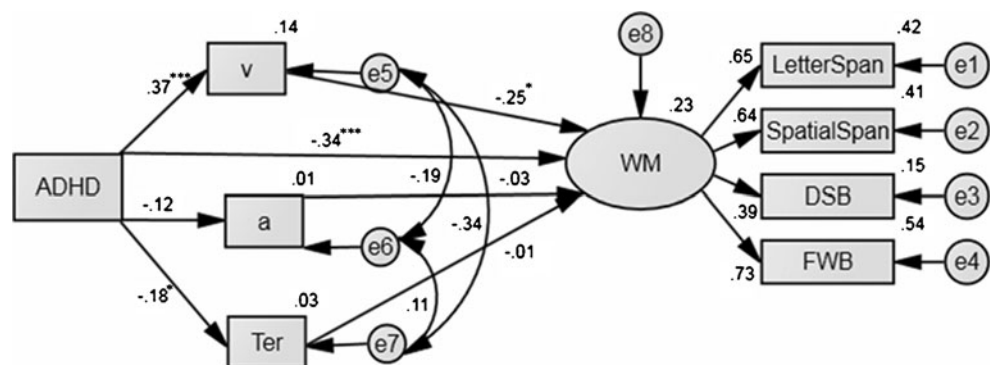
status on WM prior to the addition of the mediators was also significant ($\beta = -0.43$, $p < 0.001$).

In meditation analyses using the ex-Gaussian indicators, the same general pattern of results was found. That is, the Sobel test of the indirect effect of ADHD status on inhibitory control through τ was significant (total unstandardized indirect effect = 22.88 , $SE = 9.15$, $p = 0.006$), indicating that τ mediated the relationship between ADHD status and inhibitory control, accounting for 19 % of the variance in the path from ADHD status to SSRT. The WM model with ex-Gaussian terms as mediators was a poor fit to the data ($\chi^2[14] = 25.89$, $p = 0.03$, CFI = 0.96, RMSEA = 0.08) but τ predicted WM capacity in the expected direction (standardized regression = -0.26 , $p = 0.007$). Neither μ nor σ were mediators for either SSRT or WM variables.

Discussion

There is robust evidence that relative to non-ADHD controls, children with ADHD perform more poorly on tasks of inhibitory control and WM, and that when speeded responses are required, their performance is often characterized by slower and more variable RTs. We demonstrate that these phenomena can be integrated and understood within a diffusion model approach that isolates RTs into separable components. Specifically, children with ADHD have slower drift rates than non-ADHD controls, reflecting slower and less efficient information processing. In turn, slower drift rates partially mediate the relationship between ADHD and poor EF performance (as shown in Figs. 2 and 3). Although data is not longitudinal and causality cannot be firmly established, results lend support to recent theories that implicate a general cognitive deficit in processing efficiency that contributes to poor performance on a variety of EF tasks among children with ADHD (Alderson et al. 2007; Lijffijt et al. 2005). However, this general deficit partially mediates performance, leaving room for the possibility of specific deficits in higher-order cognitive functions or other factors to explain remaining variance.

Fig. 3 SEM demonstrating the relationships between ADHD, RT components, and WM capacity. Standardized parameter estimates displayed. *** $p < 0.001$, * $p < 0.05$



These results also help explain large between-study heterogeneity for group differences in RTs. Though slow RTs are a prominent feature in the cognitive landscape of ADHD, like EF deficits, which are found in only 30–50 % of children with ADHD (Nigg et al. 2005), slow RTs are by no means ubiquitous (Frazier et al. 2004; Huang-Pollock et al. 2012; Lijffijt et al. 2004). Using the traditional analysis approach, we found no group differences in MRT, which could have been interpreted as evidence that processing speed for children with ADHD was equivalent to non-ADHD controls. Using the diffusion model approach, however, it became clear that information processing speed/efficiency, as indexed by drift rate, is in fact slower for children with ADHD than controls.

Slower and less efficient information processing in the ADHD sample was offset by faster non-decision times (means and effect sizes for each are summarized in Table 2 under *Diffusion Modeling Components*), which indicates that children with ADHD spent less time on tasks such as stimulus encoding, motor preparation, or both. Thus, group differences in cognitive processing speed are obscured using traditional MRT analyses because children with ADHD differ from non-ADHD controls on two processes with opposing effects on the final RT. Although the effect for drift rate was stronger than that for non-decision time, unless a method of decomposing RTs is utilized, true group differences in processing speed/efficiency may not always be observed.

Similarly, when ex-Gaussian decompositions of the RT distribution were conducted, children with ADHD differed from non-ADHD controls in τ , the exponential tail of the distribution, but not in μ or σ , which characterize the normal portion of the distribution. This is consistent with previous studies demonstrating that group differences in mean and standard deviation of RT are attributable to children with ADHD having a large number of very long RTs (Buzy et al. 2009; Hervey et al. 2006; Leth-Steensen et al. 2000). Group difference in τ are consistent with the those found in drift rate because small changes in drift rates have more effect on RTs in the upper tail of the distribution than on those in the leading edge (for more detail see Wagenmakers et al. 2005). Also consistent with the fact that slow drift rates and τ have similar effects on the RT distribution, τ mediated ADHD-related deficits in SSRT and WM similarly to drift. However, the effect size for group difference in τ (as reported in Table 2) was only half of that for drift rate and the meditational model relating τ to WM was a poor fit to the data. Taken together, results indicate that in the current sample τ was a less sensitive indicator of cognitive dysfunction than drift rate. One reason for the smaller effects and weaker mediation may be that, similar to MRT, τ is impacted by multiple cognitive processes only some of which differ between groups (Matzke and Wagenmakers 2009);

however, additional studies comparing models will be critical for the field in determining how best to interpret parameters and capture the cognitive processes that are impaired in ADHD.

The finding that children with ADHD have slow drift rates is consistent with results from two other independent samples (Karalunas et al. 2012), as well as with evidence from a meta-analysis of continuous performance tasks that found slower drift rates for children with ADHD than non-ADHD controls (Huang-Pollock et al. 2012). Findings are also consistent with recent evidence that children with ADHD are less able to suppress the neurologic activation of the resting state (i.e. default-mode network activation) during active or effortful cognition (Fair et al. 2010; Fassbender et al. 2009). Single-cell recordings in non-human primates (Beck et al. 2008; Ratcliff et al. 2003) and imaging studies in humans (Heekeren et al. 2008; Ratcliff et al. 2009) both suggest that drift rate reflects the signal:noise ratio in neural circuits responsible for the decision process. When the signal:noise ratio is higher, drift rates are faster. As the signal:noise ratio decreases, drift rates become slower. Thus, difficulty suppressing resting state activation would also be expected to result in lower signal:noise ratios and slower drift rates.

Resting-state activation in ADHD has most often been described as contributing to “attention lapses,” in which the child is temporarily distracted or disengaged from the task (Sonuga-Barke and Castellanos 2007; Weissman et al. 2006). In contrast, we describe how resting-state activation may directly impact the efficiency of decision processes via decreasing signal:noise ratios (Broyd et al. 2009; Deco et al. 2009; MacDonald et al. 2009). Theories of resting-state activation hypothesize that (a) stable differences in functional connectivity and white matter integrity (Castellanos et al. 2009), and/or (b) motivational factors that impact the shift between resting and active states (Sonuga-Barke and Castellanos 2007), may both influence signal:noise ratios in information processing. The exact relationship between measures of attention, motivation, drift rate, and neural activation in resting-state networks is not clear. However, as drift rates appear to be a highly sensitive indicator of cognitive dysfunction in ADHD, future research directly addressing these relationships will be important.

As noted above, children with ADHD demonstrated *faster* non-decision times than non-ADHD controls (see group means reported in Table 2). On one hand, this could be interpreted as children with ADHD having better encoding or motor preparation than non-ADHD controls; however, this interpretation would contradict research demonstrating impaired motor function in children ADHD (Rommelse et al. 2008; Suskauer et al. 2008), and so it is important to remember that faster is not always synonymous with better. Instead, it may be that children with ADHD are

inefficient in their approach and do not spend enough time on these non-decisional processes, which may contribute to poorer overall performance. Further, although the diffusion model provides parameters with more specific cognitive interpretations than other RT models, non-decision time continues to reflect the impact of multiple cognitive processes. It is not currently possible to say which of these contributes to the group differences in this sample. Additional studies directly manipulating the complexity of motor response or correlating non-decision times to physiological indices of encoding and motor preparation will be necessary to fully understand group differences in non-decision time.

Though statistically significant, the effect size for non-decision time was small. Two previous studies, one also using a stop task, found no group differences in non-decision time (Huang-Pollock et al. 2012; Karalunas et al. 2012). Given the well-described heterogeneity in ADHD, it is possible that the effect in the current sample is driven by a small but meaningful subgroup of children with large deficits in one of the processes contributing to non-decision time. This would increase between-study heterogeneity in effects detected. Future studies that better characterize subgroups using methods such as latent class or graph theoretical approaches (Fair et al. 2012; Newman 2006) will be critical for understanding within-group heterogeneity. Similarly, our study does not address whether slow drift rates are specific to ADHD or also characterize other psychiatric disorders. Given that neither RT variability generally, τ specifically (Geurts et al. 2008), nor difficulty suppressing resting state activation (Buckholz and Meyer-Lindenberg 2012) are specific to ADHD, it is unlikely that drift rate will prove to be so. This may allow drift rate to be used as a marker of trans-diagnostic risk factors, which are of particular interest in efforts to improve definitions of psychiatric disorders (Insel et al. 2010), but future research is needed to fully confirm drift rate's sensitivity and specificity to ADHD.

In the current study, children were asked to place equal emphases on the speed and accuracy of the go process, and in this context children with ADHD did not trade speed for accuracy to a greater degree than controls (i.e. they did not differ in boundary separation), a finding which was replicated in two additional independent samples (Karalunas et al. 2012). Results suggest that choice impulsivity (e.g. prioritizing speed over accuracy due to poor task engagement or heightened desire to avoid challenging tasks, Winstanley et al. 2006) is not a primary determinant of task performance in ADHD. However, previous work found that when specifically asked to emphasize accuracy over speed, children with ADHD did trade speed for accuracy to a greater degree than non-ADHD controls (Mulder et al. 2010). Together, these findings suggest that children with

ADHD may have difficulty flexibly modulating their speed-accuracy trade-off settings to meet specific task demands. In future studies, it will be important to fully explore the degree to which children with ADHD can flexibly adjust their response style to meet task demands.

Although the current data suggest that children with and without ADHD do not differ in their speed-accuracy trade-offs in neutral instructional contexts, it is important to note how the context of the stop signal paradigm may have impacted results. Previous work in adults suggests that individuals strategically slow their responding on go trials, to minimize failures of inhibition on stop trials. This strategy results in the diffusion model detecting wider boundary separations than would be seen in a non-stop task context (Verbruggen and Logan 2009). In our sample, non-ADHD controls successfully inhibited more frequently than children with ADHD, which could suggest they adopted this strategy. The concern is that if non-ADHD controls adopt this strategy, it may be reflected in their having larger boundary separations on the stop task than they would on a non-inhibition task and could result in finding a group difference in boundary separation that resulted only from the contextual task demands. However, even in the presence of this potential confound we found no group differences in boundary separation. The demands of the stop task may also change how motor preparatory processes are carried out, impacting non-decision times. Despite concerns, the diffusion model has been applied successfully to a wide range of tasks including the stop task, with consistently good fit to the data (Ratcliff and McKoon 2008), and all participant data in our sample was well fit to the model. To mitigate the effects of stop trials on parameter estimates, we removed all stop+1 trials but it will be critical for future studies to apply the diffusion model in other choice RT paradigms to fully explore these potential confounds.

Limitations & Future Directions

The current results help elucidate mechanisms contributing to cognitive impairments in ADHD. Impairments in EF have been linked directly to poor academic achievement and problematic skill acquisition in ADHD (Biederman et al. 2004; Huang-Pollock and Karalunas 2010). Better understanding how molar EF deficits arise is therefore critically important to the development of better psychoeducational intervention strategies. Current results suggest that a common deficit in rapid, efficient decision-making accounts for a substantial proportion of variance in EF ability among children with and without ADHD. Ultimately, finding interventions to directly target slow, inefficient processing may be a more successful than targeting individual EF weaknesses. That being said, it is important to note that significant direct paths from ADHD to inhibitory control and to

WM were also found (as shown in Figs. 2 and 3, respectively). This either indicates the presence of additional mediators that were not accounted for in the model, or, because EFs are not unitary (Miyake et al. 2000), that the remaining variance reflects residual ADHD-related deficits specific to response inhibition and WM. Future studies will be required to explore these possibilities.

Several other areas will also be critical for future research to address. First, in the absence of longitudinal data or direct task manipulations of WM and inhibitory demands, it is not possible to firmly conclude that slow drift rates cause impairments in EF. The correlations between mediator and outcome could be reversed, such that poor WM and inhibition actually cause slow drift rate. Although our data cannot rule out this possibility, our interpretation of directionality rests on strong empirical evidence demonstrating that developmental improvements in WM can be attributed to development of processing speed (rather than the other way around, Fry and Hale 2000; Kail 2007) and with previous work suggesting that slow processing speed would result in poor inhibition (Alderson et al. 2007; Lijffijt et al. 2005). However, longitudinal studies with cross-lag designs that can address causality between processes that are developing in tandem (e.g. Blandon et al. 2010) or direct task manipulations will be important for fully understanding the causal relationships between these processes.

Second, diffusion model parameters were estimated from the same task that was used to estimate inhibitory control. Although the “go” and “stop” tasks that are embedded in the stop signal paradigm have been conceptualized as reflecting independent processes, estimation from the same task potentially increases the correlations between diffusion model and inhibitory control measures. However, of the diffusion model parameters, only drift rate mediated the relationship between ADHD status and inhibitory control. If mediation effects were an artifact of having estimated the diffusion parameters from the go trials of the task, then we might expect both drift rate and non-decision time to be mediators, or for all three diffusion parameters to show similar effect sizes and correlations with the SSRT outcome, which they do not. Further, drift rate also mediated the relationship between ADHD status and WM deficits. The specificity of mediation effects to a single diffusion model parameter and the replication of results across two prominent EFs suggests that results cannot simply be attributed to procedures used to estimate diffusion parameters. Nonetheless, future studies using two distinct tasks for measurement of inhibitory control and diffusion model parameters will be required.

Finally, our community recruitment procedures resulted in a sample of children with ADHD who are highly representative of the population of children with ADHD in terms of subtype composition, gender ratios, and proportion of children taking stimulant medications (Froehlich et al.

2007). Nonetheless, this resulted in the control and ADHD samples having different gender ratios. The inclusion of gender as a covariate did not affect group differences on any measure; however, samples size was not large enough to test mediational models separately for boys and girls. In future studies, it will be important to determine whether mediation effects differ by gender.

Conclusion

Overall, we present an integrated account based on a diffusion modeling technique that explains why children with ADHD often (but not always) present with slow and variable RTs, and why slower and more variable RTs regularly co-occur with impaired EF in ADHD. Such work has direct implication for prominent etiologic theories of ADHD and the search for cognitive endophenotypes, and it provides one method by which to better understand the mechanisms underlying cognitive deficits in childhood ADHD. Future work examining individual differences in the flexibility of these parameters to context, as well as their reliability across time, and relationship to neural correlates will be important next steps to assist in the elucidation of cognitive mechanisms characterizing ADHD.

Acknowledgments The project described was supported in part by R01MH084947 to Cynthia Huang-Pollock and F32MH098632 to Sarah Karalunas from the National Institutes of Mental Health. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institute of Mental Health or the National Institutes of Health.

References

- Alderson, R. M., Rapport, M. D., & Kofler, M. J. (2007). Attention-deficit/hyperactivity disorder and behavioral inhibition: a meta-analytic review of the stop-signal paradigm. *Journal of Abnormal Child Psychology*, 35(5), 745–758.
- Alderson, R. M., Rapport, M. D., Sarver, D. E., & Kofler, M. J. (2008). Adhd and behavioral inhibition: a re-examination of the stop-signal task. [Journal; Peer Reviewed Journal]. *Journal of Abnormal Child Psychology: An official publication of the International Society for Research in Child and Adolescent Psychopathology*, 36(7), 989–998.
- Balota, D. A., & Yap, M. J. (2011). Moving beyond the mean in studies of mental chronometry: the power of response time distributional analyses. *Current Directions in Psychological Science*, 20(3), 160–166.
- Banaschewski, T., Yordanova, J., Kolev, V., Heinrich, H., Albrecht, B., & Rothenberger, A. (2008). Stimulus context and motor preparation in attention-deficit/hyperactivity disorder. *Biological Psychology*, 77(1), 53–62.
- Beck, J. M., Ma, W. J., Kiani, R., Hanks, T., Churchland, A. K., Roitman, J., Shadlen, M. N., Latham, P. E., & Pouget, A. (2008). Probabilistic population codes for bayesian decision making. *Neuron*, 60(6), 1142–1152.

- Biederman, J., Monuteaux, M. C., Doyle, A. E., Seidman, L. J., Wilens, T. E., Ferrero, F., Morgan, C. L., & Faraone, S. V. (2004). Impact of executive function deficits and attention-deficit/hyperactivity disorder (adhd) on academic outcomes in children. *Journal of Consulting and Clinical Psychology*, 72(5), 757–766.
- Blandon, A. Y., Calkins, S. D., Grimm, K. J., Keane, S. P., & O'Brien, M. (2010). Testing a developmental cascade model of emotional and social competence and early peer acceptance. *Development and Psychopathology*, 22, 737–748.
- Bogacz, R., Brown, E., Moehlis, J., Holmes, P., & Cohen, J. D. (2006). The physics of optimal decision making: a formal analysis of models of performance in two-alternative forced-choice tasks. *Psychological Review*, 113(4), 700–765.
- Broyd, S. J., Demanuele, C., Debener, S., Helps, S. K., James, C. J., & Sonuga-Barke, E. (2009). Default-mode brain dysfunction in mental disorders: a systematic review. *Neuroscience and Biobehavioral Reviews*, 33, 279–296.
- Buckholtz, J. W., & Meyer-Lindenberg, A. (2012). Psychopathology and the human connectome: toward a transdiagnostic model of risk for mental illness. *Neuron*, 74(6), 990–1004.
- Buzy, W. M., Medoff, D. R., & Schweitzer, J. B. (2009). Intra-individual variability among children with adhd on a working memory task: an ex-gaussian approach. *Child Neuropsychology*, 15(5), 441–459.
- Castellanos, F. X., & Tannock, R. (2002). Neuroscience of attention-deficit/hyperactivity disorder: the search for endophenotypes. *Nature Reviews Neuroscience*, 3(8), 617–628.
- Castellanos, F. X., Sonuga-Barke, E. J. S., Scheres, A., Di Martino, A., Hyde, C., & Walters, J. R. (2005). Varieties of attention-deficit/hyperactivity disorder-related intra-individual variability. *Biological Psychiatry*, 57(11), 1416–1423.
- Castellanos, F. X., Kelly, C., & Milham, M. P. (2009). The restless brain: attention-deficit hyperactivity disorder, resting-state functional connectivity, and intrasubject variability. *Canadian Journal of Psychiatry-Revue Canadienne De Psychiatrie*, 54(10), 665–672.
- Conners, C. K. (2001). *Conners' rating scales—revised technical manual*. North Tonawanda, NY: Multi-Health Systems Inc.
- Conway, A., Kane, M. J., Bunting, M. F., Hambrick, D. Z., Wilhelm, O., & Engle, R. W. (2005). Working memory span tasks: a methodological review and user's guide. *Psychonomic Bulletin & Review*, 12(5), 769–786.
- Deco, G., Jirsa, V., McIntosh, A. R., Sporns, O., & Kotter, R. (2009). Key role of coupling, delay, and noise in resting brain fluctuations. *PNAS Proceedings of the National Academy of Sciences of the United States of America*, 106(25), 10302–10307.
- Douglas, V. I. (1999). Cognitive control processes in attention-deficit/hyperactivity disorder. In H. C. Quay & A. E. Hogan (Eds.), *Handbook of disruptive behavior disorders* (Vol. xiii) (pp. 105–138). NY: Kluwer Academic Publishers.
- Epstein, J. N., Langberg, J. M., Rosen, P. J., Graham, A., Narad, M. E., Antonini, T. N., Brinkman, W. B., Froehlich, T., Simon, J. O., & Altaye, M. (2011). Evidence for higher reaction time variability for children with adhd on a range of cognitive tasks including reward and event rate manipulations. *Neuropsychology*, 25(4), 427–441.
- Fair, D. A., Posner, J., Nagel, B. J., Bathula, D., Costa Dias, T. G., Mills, K. L., Blythe, M. S., Giwa, A., Schmitt, C., & Nigg, J. T. (2010). Atypical default network connectivity in youth with attention-deficit/hyperactivity disorder. *Biological Psychiatry*, 68(12), 1084–1091.
- Fair, D., Bathula, D., Nikolas, M., & Nigg, J. T. (2012). Distinct neuropsychological subgroups in typically developing youth inform heterogeneity in children with adhd. *Proceedings of the National Academy of Sciences of the United States of America*. doi:10.1073/pnas.1115365109.
- Faraone, S. V., Perlis, R. H., Doyle, A., Smoller, J. W., Goralnick, J. J., Holmgren, M. A., & Sklar, P. (2005). Molecular genetics of attention-deficit/hyperactivity disorder. *Biological Psychiatry*, 57(11), 1313–1323.
- Fassbender, C., Zhang, H., Buzy, W. M., Cortese, C. R., Mizuiri, D., Beckett, L., & Schweitzer, J. B. (2009). A lack of default network suppression is linked to increased distractibility in adhd. *Brain Research*, 1273(1), 114–128.
- Frazier, T. W., Demaree, H. A., & Youngstrom, E. A. (2004). Meta-analysis of intellectual and neuropsychological test performance in attention-deficit/hyperactivity disorder. *Neuropsychology*, 18(3), 543–555.
- Froehlich, T. E., Lanphear, B., Epstein, J. N., Barbaresi, W. J., Katusic, S. K., & Kahn, R. S. (2007). Prevalence, recognition, and treatment of attention-deficit/hyperactivity disorder in a national sample of us children. *Archives of Pediatrics & Adolescent Medicine*, 161(9), 857–864.
- Fry, A. F., & Hale, S. (2000). Relationships among processing speed, working memory, and fluid intelligence in children. *Biological Psychology*, 54(1–3), 1–34.
- Geurts, H. M., Grasman, R. P. P., Verté, S., Oosterlaan, J., Roeyers, H., van Kammen, S. M., & Sergeant, J. A. (2008). Intra-individual variability in adhd, autism spectrum disorders and tourette's syndrome. *Neuropsychologia*, 46(13), 3030–3041.
- Grudnik, J. L., & Kranzler, J. H. (2001). Meta-analysis of the relationship between intelligence and inspection time. *Intelligence*, 29(6), 523–535.
- Halperin, J. M., & Schulz, K. P. (2006). Revisiting the role of the prefrontal cortex in the pathophysiology of attention-deficit/hyperactivity disorder. *Psychological Bulletin*, 132(4), 560–581.
- Heekeren, H. R., Marrett, S., & Ungerleider, L. G. (2008). The neural systems that mediate human perceptual decision making. *Nature Reviews Neuroscience*, 9(6), 467–479.
- Hervey, A. S., Epstein, J. N., Curry, J. F., Tonev, S., Arnold, L. E., Conners, C. K., Hinshaw, S. P., Swanson, J. M., & Hechtman, L. (2006). Reaction time distribution analysis of neuropsychological performance in an adhd sample. *Child Neuropsychology*, 12(2), 125–140.
- Huang-Pollock, C. L., & Karalunas, S. L. (2010). Working memory demands impair skill acquisition in children with adhd. *Journal of Abnormal Psychology*, 119(1), 174–185.
- Huang-Pollock, C. L., Karalunas, S. L., Moore, A., & Tam, H. (2011). Working memory subsystems and their contribution to working memory deficits in children with adhd. Poster presented at the biennial meeting of the International Society for Research in Child and Adolescent Psychopathology, Chicago, IL.
- Huang-Pollock, C. L., Karalunas, S. L., Tam, H., & Moore, A. N. (2012). Evaluating vigilance deficits in adhd: a meta-analysis of cpt performance. *Journal of Abnormal Psychology*, 121(2), 360–371.
- Hurks, P. P. M., Adam, J. J., Hendriksen, J. G. M., Vles, J. S. H., Feron, F. J. M., Kalff, A. C., Kroes, M., Steyaert, J., Crolla, I. F. A. M., van Zeben, T. M. C. B., & Jolles, J. (2005). Controlled visuomotor preparation deficits in attention-deficit/hyperactivity disorder. *Neuropsychology*, 19(1), 66–76.
- Insel, T., Cuthbert, B., Garvey, M., Heinssen, R., Pine, D. S., Quinn, K., Sanislow, C., & Wang, P. (2010). Research domain criteria (rdc): toward a new classification framework for research on mental disorders. *The American Journal of Psychiatry*, 167(7), 748–751.
- Kail, R. (2007). Longitudinal evidence that increases in processing speed and working memory enhance children's reasoning. *Psychological Science*, 18(4), 312–313.
- Karalunas, S. L., Huang-Pollock, C. L., & Nigg, J. T. (2012). Decomposing adhd-related effects in response speed and variability. *Neuropsychology*, 26(6), 684–694.
- Klimkeit, E. I., Mattingley, J. B., Sheppard, D. M., Lee, P., & Bradshaw, J. L. (2005). Motor preparation, motor execution,

- attention, and executive functions in attention deficit/hyperactivity disorder (adhd). *Child Neuropsychology*, 11(2), 153–173.
- Kühn, S., Schmiedek, F., Schott, B., Ratcliff, R., Heinze, H.-J., Düzcel, E., Lindenberger, U., & Lövdén, M. (2011). Brain areas consistently linked to individual differences in perceptual decision-making in younger as well as older adults before and after training. *Journal of Cognitive Neuroscience*, 23(9), 2147–2158.
- Kuntsi, J., & Stevenson, J. (2001). Psychological mechanisms in hyperactivity: Is the role of genetic factors. *Journal of Child Psychology and Psychiatry*, 42(2), 211–219.
- Kuntsi, J., Rogers, H., Swinard, G., Borger, N., van der Meere, J., Rijdsdijk, F., & Asherson, P. (2006). Reaction time inhibition, working memory and 'delay aversion' performance: genetic influences and their interpretation. *Psychological Medicine*, 36(11), 1613–1624.
- Lacouture, Y., & Cousineau, D. (2008). How to use matlab to fit the ex-gaussian and other probability functions to a distribution of response times. *Tutorials in Quantitative Methods for Psychology*, 4(1), 35–45.
- Lahey, B. B., Applegate, B., McBurnett, K., Biederman, J., Greenhill, L., Hynd, G. W., Barkley, R. A., Newcorn, J., Jensen, P., Richters, J., Garfinkel, B., Kerdyk, L., Frick, P. J., Ollendick, T., Perez, D., Hart, E. L., Waldman, I., & Shaffer, D. (1994). Dsm-iv field trials for attention-deficit hyperactivity disorder in children and adolescents. *The American Journal of Psychiatry*, 151(11), 1673–1685.
- Lajoie, G., Anderson, V., Anderson, P., Tucker, A. R., Robertson, I. H., & Manly, T. (2005). Effects of methylphenidate on attention skills in children with attention deficit/hyperactivity disorder. *Brain Impairment*, 6(1), 21–32.
- Leth-Steensen, C., Elbaz, Z. K., & Douglas, V. I. (2000). Mean response times, variability and skew in the responding of adhd children: a response time distributional approach. *Acta Psychologica*, 104(2), 167–190.
- Lijffijt, M., Bekker, E. M., Quik, E. H., Bakker, J., Kenemans, J. L., & Verbaten, M. N. (2004). Differences between low and high trait impulsivity are not associated with differences in inhibitory motor control. *Journal of Attention Disorders*, 8(1), 25–32.
- Lijffijt, M., Kenemans, J. L., Verbaten, M. N., & van Engeland, H. (2005). A meta-analytic review of stopping performance in attention-deficit/hyperactivity disorder: deficient inhibitory motor control? *Journal of Abnormal Psychology*, 114(2), 216–222.
- MacDonald, S. W. S., Li, S.-C., & Bäckman, L. (2009). Neural underpinnings of within-person variability in cognitive functioning. *Psychology and Aging*, 24(4), 792–808.
- Martinussen, R., Hayden, J., Hogg-Johnson, S., & Tannock, R. (2005). A meta-analysis of working memory impairments in children with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child and Adolescent Psychiatry*, 44(4), 377–384.
- Matzke, D., & Wagenmakers, E.-J. (2009). Psychological interpretation of the ex-gaussian and shifted wald parameters: a diffusion model analysis. *Psychonomic Bulletin & Review*, 16(5), 798–817.
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: a latent variable analysis. *Cognitive Psychology*, 41(1), 49–100.
- Mulder, M. J., Bos, D., Weusten, J. M. H., van Belle, J., van Dijk, S. C., Simen, P., van Engeland, H., & Durston, S. (2010). Basic impairments in regulating the speed-accuracy tradeoff predict symptoms of attention-deficit/hyperactivity disorder. *Biological Psychiatry*, 68(12), 1114–1119.
- Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences of the United States of America*, 103(23), 8577–8582.
- Nigg, J. T., Willcutt, E. G., Doyle, A., & Sonuga-Barke, E. J. S. (2005). Causal heterogeneity in attention-deficit/hyperactivity disorder: do we need neuropsychologically impaired subtypes? *Biological Psychiatry*, 57(11), 1224–1230.
- Philastides, M. G., & Sajda, P. (2006). Temporal characterization of the neural correlates of perceptual decision making in the human brain. *Cerebral Cortex*, 16(4), 509–518.
- Philastides, M. G., Ratcliff, R., & Sajda, P. (2006). Neural representation of task difficulty and decision making during perceptual categorization: a timing diagram. *The Journal of Neuroscience*, 26(35), 8965–8975.
- Preacher, K. J., & Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, 40, 879–891.
- Ratcliff, R. (2002). A diffusion model account of response time and accuracy in a brightness discrimination task: fitting real data and failing to fit fake but plausible data. *Psychonomic Bulletin & Review*, 9(2), 278–291.
- Ratcliff, R., & McKoon, G. (2008). The diffusion decision model: theory and data for two-choice decision tasks. *Neural Computation*, 20(4), 873–922.
- Ratcliff, R., & Rouder, J. N. (1998). Modeling response times for two-choice decisions. *Psychological Science*, 9(5), 347–356.
- Ratcliff, R., Cherian, A., & Segraves, M. (2003). A comparison of macaque behavior and superior colliculus neuronal activity to predictions from models of two-choice decisions. *Journal of Neurophysiology*, 90(3), 1392–1407.
- Ratcliff, R., Thapar, A., Gomez, P., & McKoon, G. (2004). A diffusion model analysis of the effects of aging in the lexical-decision task. *Psychology and Aging*, 19(2), 278–289.
- Ratcliff, R., Philastides, M. G., & Sajda, P. (2009). Quality of evidence for perceptual decision making is indexed by trial-to-trial variability of the eeg. *PNAS Proceedings of the National Academy of Sciences of the United States of America*, 106(16), 6539–6544.
- Reynolds, C., & Kamphaus, R. (2004). *Behavioral assessment system for children, 2nd ed. Manual*. Circle Pines, MN: AGS Publishing.
- Rommelse, N. N., Altink, M. E., Oosterlaan, J., Beem, L., Buschgens, C. J. M., Buitelaar, J. K., & Sergeant, J. A. (2008). Speed, variability, and timing of motor output in adhd: which measures are useful for endophenotypic research. *Behavior Genetics*, 38(2), 121–132.
- Salthouse, T. A., McGuthry, K. E., & Hambrick, D. Z. (1999). A framework for analyzing and interpreting differential aging patterns: application to three measures of implicit learning. [Article]. *Aging Neuropsychology and Cognition*, 6(1), 1–18.
- Sattler, J. (2008). *Resource guide to accompany assessment of children: Cognitive foundations* (5th ed.). San Diego: Jerome Sattler Publisher, Inc.
- Schachar, R., Chen, S., Logan, G., Ornstein, T. J., Crosbie, J., Ickowicz, A., & Pakulak, A. (2004). Evidence for an error monitoring deficit in attention deficit hyperactivity disorder. *Journal of Abnormal Child Psychology*, 32(3), 285–293.
- Schmiedek, F., Oberauer, K., Wilhelm, O., Süß, H.-M., & Wittmann, W. W. (2007). Individual differences in components of reaction time distributions and their relations to working memory and intelligence. *Journal of Experimental Psychology: General*, 136(3), 414–429.
- Schmiedek, F., Lövdén, M., & Lindenberger, U. (2009). On the relation of mean reaction time and intraindividual reaction time variability. *Psychology and Aging*, 24(4), 841–857.
- Sergeant, J. (2000). The cognitive-energetic model: an empirical approach to attention-deficit hyperactivity disorder. *Neuroscience and Biobehavioral Reviews*, 24(1), 7–12.
- Sergeant, J. A., & Scholten, C. A. (1985). On data limitations in hyperactivity. *Journal of Child Psychology and Psychiatry*, 26(1), 111–124.
- Sergeant, J., Oosterlaan, J., & van der Meere, J. (1999). Information processing and energetic factors in attention deficit/hyperactivity disorder. In H. Quay & A. Hogan (Eds.), *Handbook of disruptive behavior disorders*. NY: Kluwer Academic Publishers.
- Sheslow, D., & Adams, W. (2003). *Wide range assessment of memory and learning, 2nd ed (wraml-2): Administration and technical manual*. Wilmington, DE: Wide Range.

- Sonuga-Barke, E., & Castellanos, F. X. (2007). Spontaneous attentional fluctuations in impaired states and pathological conditions: a neurobiological hypothesis. *Neuroscience and Biobehavioral Reviews*, 31(7), 977–986.
- Spaniol, J., & Bayen, U. J. (2005). Aging and conditional probability judgments: a global matching approach. *Psychology and Aging*, 20(1), 165–181.
- Steger, J., Imhof, K., Coutts, E., Gundelfinger, R., Steinhausen, H. C., & Brandeis, D. (2001). Attentional and neuromotor deficits in adhd. *Developmental Medicine and Child Neurology*, 43(3), 172–179.
- Suskauer, S. J., Simmonds, D. J., Caffo, B. S., Denckla, M. B., Pekar, J. J., & Mostofsky, S. H. (2008). Fmri of intrasubject variability in adhd: anomalous premotor activity with prefrontal compensation. *Journal of the American Academy of Child and Adolescent Psychiatry*, 47(10), 1141–1150.
- Thapar, A., Ratcliff, R., & McKoon, G. (2003). A diffusion model analysis of the effects of aging on letter discrimination. *Psychology and Aging*, 18(3), 415–429.
- Verbruggen, F., & Logan, G. D. (2009). Proactive adjustments of response strategies in the stop-signal paradigm. *Journal of Experimental Psychology: Human Perception and Performance*, 35(3), 835–854.
- Verbruggen, F., Logan, G., Liefoghe, B., & Vandierendonck, A. (2008). Short-term aftereffects of response inhibition: repetition priming or between-trial control adjustments. *Journal of Experimental Psychology of Human Performance*, 34(2), 413–426.
- Voss, A., & Voss, J. (2007). Fast-dm: a free program for efficient diffusion model analysis. *Behavior Research Methods*, 39(4), 767–775.
- Voss, A., Rothermund, K., & Voss, J. (2004). Interpreting the parameters of the diffusion model: an empirical validation. *Memory & Cognition*, 32(7), 1206–1220.
- Wagenmakers, E.-J., Grasman, R. P. P. P., & Molenaar, P. C. M. (2005). On the relation between the mean and the variance of a diffusion model response time distribution. *Journal of Mathematical Psychology*, 49, 195–204.
- Wechsler, D. (2003). *Wechsler intelligence scale for children, 4th ed (wisc-iv) technical and interpretive manual*. San Antonio: Harcourt Brace.
- Weissman, D. H., Roberts, K. C., Visscher, K. M., & Woldorff, M. G. (2006). The neural bases of momentary lapses in attention. *Nature Neuroscience*, 9, 971–978.
- Widaman, K. F. (2006). Best practices in quantitative methods for developmentalists: Iii. Missing data: what to do with or without them. *Monographs of the Society for Research in Child Development*, 71(3), 42–64.
- Willcutt, E. G., Doyle, A., Nigg, J. T., Faraone, S. V., & Pennington, B. F. (2005). Validity of the executive function theory of attention-deficit/hyperactivity disorder: a meta-analytic review. *Biological Psychiatry*, 57(11), 1336–1346.
- Winstanley, C. A., Eagle, D. M., & Robbins, T. W. (2006). Behavioral models of impulsivity in relation to adhd: translation between clinical and preclinical studies. *Clinical Psychology Review*, 26(4), 379–395.